

SA is switched on to Mahlobo's wheeling and dealing in nuclear bid

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President Zuma and energy minister Mahlobo are so desperate to restart the process to procure a fleet of new nuclear power stations that they are pressuring the department of energy (DoE) to produce a new "policy adjusted" electricity plan, which overturns the conclusions of its own computer modelers that nuclear energy is not needed.

The outputs from the world-class PLEXOS model, which has been run for the DoE so that it can publish an updated electricity integrated resource plan (IRP), clearly show that the energy mix that will provide South Africa with an adequate and reliable electricity supply, at least-cost, excludes nuclear energy.

Much has changed since the official IRP 2010 was published. Electricity demand is lower than a decade ago and the most optimistic new forecasts assume much slower increases in electricity demand, even if South Africa's economy expands more rapidly.

The relative costs of power generation options have also changed. The updated model incorporates latest cost data, including actual solar and wind energy prices contracted in South Africa's renewable energy IPP programme. The costs of nuclear energy used in the model were supplied by the DoE.

I could quibble with the assumptions of the modelers. Their electricity demand forecasts are almost certainly too high, as are their assumptions on renewable energy costs. This year, in a number of emerging economies, we have seen solar and wind auctions deliver prices that are half those registered in the last bid round in South Africa. And the DoE's nuclear cost assumptions seem much lower than many recent contracts, such as the Rosatom deal in Egypt.

But even with these conservative assumptions, nuclear energy is not picked by the cost-optimising model in any scenario other than one where artificial constraints are placed on how much solar and wind energy can be built and where additional carbon budget limits are imposed. Even in this extreme scenario, nuclear energy would only be required after 2039.

These outcomes are consistent with the IRP update undertaken by the DoE in 2016 under the direction of the Ministerial Advisory Council on Energy and with the CSIR's model which used the same PLEXOS software and assumptions as the DoE. A few years back, the National Planning Commission asked South Africa's premier university-based energy modeling group – the University of Cape Town's Energy Research Centre – to run an independent model. They too came to the same conclusion.

These outcomes have not been welcomed by the nuclear section in DoE and the modelers were asked to run a scenario where 9600MW of nuclear power stations are “hardwired” or forced into the model. Fortunately the modelers also calculated the additional cost of electricity in this scenario – R800 billion over 20 years!

Originally, nuclear proponents relied on the IRP 2010 for their justification for starting the procurement of new power stations. Now that the updated IRPs indicate we don’t need nuclear energy, they question the usefulness of the models (even with their conservative input assumptions) and instead emphasize the need for a “policy adjusted” plan that takes into account other objectives such as job creation, local manufacture, use of local energy resources and reduced carbon emissions and water usage. But other technologies fare even better than nuclear in meeting these additional criteria.

The optimal, least-cost mix recommended by all these models includes solar and wind energy complemented by pumped storage, hydro-electricity and gas turbines or engines, on top of legacy coal fired powered stations.

But, cry the nuclear advocates, solar and wind energy are intermittent, we cannot rely on them. However, the planning models fully account for their variability and pick a least-cost energy mix that meets specified reliability criteria at any time of the day or year. Indeed, the modelers assert that their optimal mix includes sufficient dispatchable power even when the sun doesn’t shine and the wind doesn’t blow.

But, says our energy minister, our energy security depends on energy diversity – we need to develop all sources of energy. I have never understood this line of argument. What kind of policy suggests that we do everything; use all of the above? Good policy requires informed choices. We clearly need to opt for the lowest cost power sources that offer us adequate power to grow our economy and improve the welfare of all our people whilst still meeting broader social and environmental goals.

As arguments run out for nuclear, the diehard proponents point to Denmark and Germany which have high shares of renewable energy and high electricity prices, conveniently omitting that these countries were early movers when solar and wind energy were still expensive and that they created large new markets which catalysed higher production volumes and innovation which led to price reductions from which we can now benefit.

There are no sound arguments for building nuclear power plants. South Africa does not need them, they are too expensive and they will be difficult to finance. Even if Russia or China offer financing terms, National Treasury will have to provide sovereign guarantees to back the loan agreements and there is no fiscal space left for this. With no arguments left, those who continue to advocate restarting a nuclear programme need to demonstrate that their motives are not corrupt.

A judgement of the Western Cape High Court earlier this year made clear that the energy minister will need to make a new determination in terms of the Electricity Regulation Act if he wishes to procure nuclear energy and this should be based on a valid electricity integrated resource plan. The Minister also needs to obtain the concurrence of the National

Energy Regulator of South Africa who in turn has to first undertake a thorough public consultation process.

The minister is trying to rush through the first step by doctoring the IRP update. He will be challenged in court. The subsequent steps will also be scrutinized by civil society organisations.

Zuma has fired four previous energy ministers for not delivering his nuclear programme. I do not easily celebrate government incompetence, but in this instance I do. Minister Mahlobo is demonstrating more energy and determination, but he too will fail. What a waste of time and resources!

Nevertheless, our energy future is bright. South Africa has amongst the best solar and wind resources in the world. We have the institutions, capabilities, experience and private finance to capitalize on these new low cost technologies, including complementary, flexible power resources, storage and demand-side management. We also have the minerals and rare earths that the manufacture of these technologies require, and South Africa can grow new exports. The changes and innovation disrupting the power sector are unprecedented and likely irresistible. In a few years no one will even remember Zuma and Mahlobo's vain efforts to subvert our energy future.

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